AI ENGINEERING TEAM · DECEMBER 2025

AIRLINE SENTIMENT ANALYSIS

Business Value Document

Prepared for Executive Leadership | DistilBERT NLP Model v13 | December 2025

01 The Bottom Line

Financial & Operational Summary

Converting 14,604 daily tweets from unstructured digital noise into real-time operational intelligence – at **86.00% accuracy** and **0.08s response time** – eliminating 8,000+ hours of annual manual complaint classification.

Investment vs. Return – What We Spent vs. What We Save

~$18K Initial Investment Development cost + Infrastructure	**>$240K** Projected Annual Savings Replacing 8,000+ hours of manual work	**~1,200%** Expected ROI Within the first year
86.00% Classification Accuracy Exceeded the 75% target threshold	**0.08 sec** Inference Speed Optimized for real-time monitoring	**14,604** Tweets Analyzed Across 6 major airlines

Project Status – Current Snapshot

Status	Green Light Model promoted to Production (v13) via MLflow
Goals	All technical benchmarks met and exceeded
Timeline	Delivered on schedule – within a 4-week development sprint
Current Phase	Phase 1 complete – ready to enter Phase 2

02 Strategic Context

Why This Project, Why Now?

Market Need – The Problem We're Solving

Airlines receive thousands of social media mentions every day. 62.72% of those are negative, and 4 out of 5 complaints are about human service interactions – not operations. The problem is not the aircraft; it's the frontline staff and escalation protocols.

Without the Solution	With the Solution
• Teams manually classify every complaint • Responses delayed by hours or days • Negative content spreads unchecked • Management lacks competitor benchmarking data • US Airways: 77.7% negative – zero automated intervention	• Instant automated classification – zero manual effort • Real-time alerts for critical complaints • Live management monitoring dashboard • Visual airline performance benchmarking • Targeted response within under one hour

Strategic Alignment – How This Serves Company Goals

Goal	Business Impact
Employee Efficiency	Frees monitoring teams from manual classification so they focus on response and resolution
Data-Driven Decisions	Management receives live reports instead of weekly manual summaries
Competitive Advantage	Southwest leads with 23.6% positive sentiment – we now have a measurable roadmap to match it
Response Speed	Automated alert system ensures critical complaints are addressed within one hour

03 Impact & Outcomes

Real-World Results

Accuracy in Business Terms

The model classifies nearly 9 out of every 10 tweets correctly (86%). Critical complaints (negative sentiment) are detected at 91% accuracy – meaning 91 out of every 100 angry tweets are automatically flagged for immediate action.

91%	72%	81%
Negative Complaints Detection accuracy for critical complaints	Neutral Tweets Improved via Extended Epochs and Early Stopping	Positive Tweets Significant improvement in Version 13

Efficiency Gains – Time & Effort Savings

Task	Without Tool	With Tool
Classifying 14,604 tweets	~730 hours	< 30 minutes
Detecting a critical complaint	4–24 hours	< 0.08 seconds
Airline comparison report	1 full week	Instant – Live Dashboard
Complaint pattern analysis	Not available	Automated + Categorized

Key Business Findings

Human service is the #1 complaint driver: 31.7% of all negative tweets – not flight delays or lost baggage

United Airlines faces a reputation crisis: 68.9% negative sentiment – requires urgent intervention

US Airways is the worst performer: 77.7% negative – the highest rate in the sector

Southwest is the benchmark to follow: 23.6% positive – a replicable blueprint for improvement

Peak complaint window is 6–9 AM: Volume surges from 131 tweets at midnight to 999 by 9 AM

Negative content spreads faster: Retweet rate 0.093 vs. 0.070 for positive – a real viral risk

04 Project Roadmap & Milestones

Timeline & Delivery

Phase	Timeline	Activity	Status
Phase 0	Completed (Dec 2025)	Model development – Fine-tune DistilBERT v13 and promote to Production via MLflow	Done
Phase 1	Month 1 (Jan 2026)	Deploy FastAPI + Streamlit + configure automated alerts for United and US Airways accounts	In Progress
Phase 2	Months 1–3 (Q1 2026)	CRM integration – route high-severity complaints directly to Service Recovery teams	Scheduled
Phase 3	Months 3–6 (Q2 2026)	Expand to Instagram and Facebook + multilingual model deployment	Planned

Go-to-Market Date: The tools will be available to monitoring teams within the first month of Phase 1 approval. Customer service teams will begin receiving intelligent alerts from Day 1.

05 Risks & Mitigation

We Have a Plan B

Risk	Potential Impact	Mitigation Plan
Evolving tweet language & context drift	Accuracy degrades over time without updates	Plan: Quarterly retraining with 5,000+ fresh tweets
High GPU server costs	Real-time inference halted if budget is exceeded	Alternative: AWS/GCP Spot Instances – up to 70% cost reduction
Neutral class accuracy gap (72%)	Errors on ambiguous mixed-sentiment tweets	Plan: Build ensemble model in Phase 3 (BERT + LSTM). **Partially addressed in V13 via Class Weights and dynamic Patience control.**
Geographic data bias (Boston)	33.4% of data from Boston – regional skew	Plan: Expand dataset with national coverage in Phase 2

06 The Call to Action

What We Need From You

We need your decisions on three fronts to move from Phase 1 into Phase 2 and scale fully:

1. **Approve GPU server budget for production** — URGENT
   ~$800–1,200/month (AWS/GCP) to sustain 0.08s inference in the production environment

2. **Authorize hiring 4-6 social monitoring agents** — URGENT
   Shift: 6 AM – 12 PM (peak hours) to act on automated alerts in real time

3. **Provide direction on rolling out the tool to other departments** — MEDIUM
   Should we expand to internal analytics – employee reviews, flight satisfaction surveys?

> *Airlines that invest in frontline staff empowerment and proactive communication will differentiate themselves in a market where customer expectations are high but satisfaction is consistently low. Southwest Airlines provides a replicable blueprint – and this system gives every airline the data infrastructure needed to follow it.*